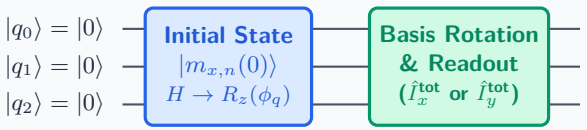
Quantum Circuit Architecture for 1D NMR Spectroscopy

Qiskit Fall Fest 2026 – HQS Quantum Simulations Challenge | State Preparation, Trotterized Time Evolution & Transverse Readout

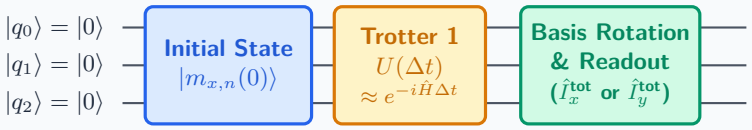
1. Time-Stepped Quantum Circuit Pipeline ($t_k = k \cdot \Delta t$)

Projective measurement collapses the quantum state. Consequently, each discrete time point $t_k = k \cdot \Delta t$ ($k = 0, \ldots, K-1$) is evaluated via a dedicated circuit containing k consecutive Trotter blocks, followed by a single terminal readout:

Step $k = 0$ ($t_0 = 0$ s):
(0 Trotter blocks, weight $m_x(0)$)



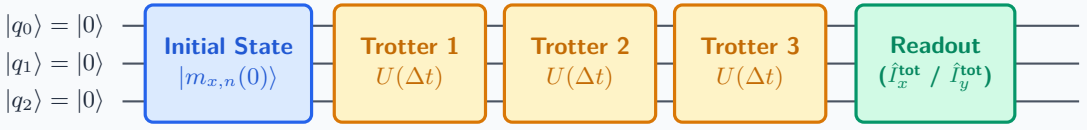
Step $k = 1$ ($t_1 = \Delta t$):
(1 Trotter block, $t_1 = 2.5$ ms)



Step $k = 2$ ($t_2 = 2\Delta t$):
(2 Trotter blocks, $t_2 = 5.0$ ms)



Step $k = 3$ ($t_3 = 3\Delta t$):
(3 Trotter blocks, $t_3 = 7.5$ ms)



$\vdots$ **(repeated for all steps up to $k = K-1 = 299$, i.e., $t_{299} = 0.7475$ s)** $\vdots$

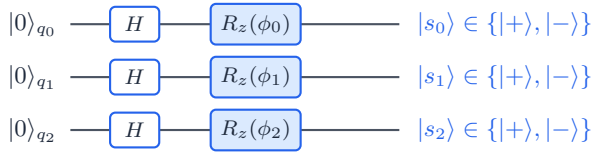
Step $k = 299$ (t_{299}):
(299 Trotter blocks in series)



2. Gate-Level Decomposition of Circuit Modules

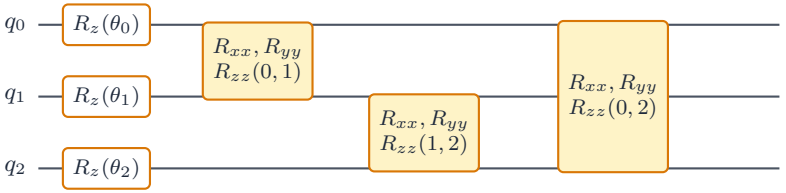
A) Initial State Preparation (`qc_init_template`)

Prepares X-eigenstates $|m_{x,n}(0)\rangle \in \{|+\rangle, |-\rangle\}^{\otimes N}$ ($\phi_q = 0 \to |+\rangle$, $\phi_q = \pi \to -i|-\rangle$):



B) Single Trotter Evolution Step $U(\Delta t)$ (`PauliEvolutionGate`)

Rotating-frame evolution $\hat{H}_{\text{rot}} = \sum_\ell \Delta\omega_\ell \hat{I}_\ell^z + 2\pi \sum_{k<\ell} J_{k\ell}(\hat{I}_k^x \hat{I}_\ell^x + \hat{I}_k^y \hat{I}_\ell^y + \hat{I}_k^z \hat{I}_\ell^z)$:



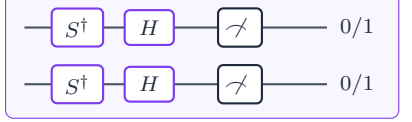
C) Transverse Magnetization Readout (X- and Y-Basis)

Hardware natively measures in the Z-basis ($|0\rangle/|1\rangle$). Since $[\hat{I}_x^{\text{tot}}, \hat{I}_y^{\text{tot}}] \neq 0$, two circuit executions with distinct pre-measurement basis rotations are required per step:

Circuit 1: $\langle \hat{I}_x^{\text{tot}}(t_k) \rangle$ (`obs_x`)



Circuit 2: $\langle \hat{I}_y^{\text{tot}}(t_k) \rangle$ (`obs_y`)



- **X-Basis Readout:** Hadamard gate H maps $|+\rangle \to |0\rangle$ and $|-\rangle \to |1\rangle$.
- **Y-Basis Readout:** $S^\dagger$ followed by H (equivalent to $R_x(\pi/2)$) maps $|+i\rangle \to |0\rangle$ and $|-i\rangle \to |1\rangle$.
- **Implementation:** Qiskit's `Estimator` automatically appends these basis-change gates.

Computational Scaling & Hardware Execution Time (`ibm_marrakesh`)

Qiskit Fall Fest 2026 – HQS Challenge | Exact Resource Scaling for $K = 300, S = 1\,024$ vs. $K = 50, S = 1\,000$ Across Molecules A, B & C

1. General Scaling Formulas

System and simulation parameters:

- N: Number of nuclear spins / qubits ($N = 2$ for Molecule C, $N = 3$ for Molecules A & B)
- K: Number of discrete time steps ($k = 0, \ldots, K-1$; baseline $K = 300$, NISQ $K = 50$)
- M: Number of initialized X-eigenstates $|m_{x,n}(0)\rangle$:

$$M_{\text{full}} = 2^N \quad \text{vs.} \quad M_{\text{sym}} = \begin{cases} 2^{N-1} & (N \text{ odd}) \\ \frac{1}{2}\left(2^N - \binom{N}{N/2}\right) & (N \text{ even}) \end{cases}$$

- Factor of $\mathbf{2}$: Separate circuit executions for $\langle \hat{I}_x^{\text{tot}}(t_k) \rangle$ and $\langle \hat{I}_y^{\text{tot}}(t_k) \rangle$
- S: Number of measurement shots per circuit ($S = 1\,024$ or $S = 1\,000$)

A) Total Number of Distinct Quantum Circuits (N_{circuits}):

$$\mathbf{N_{\text{circuits}} = 2 \cdot M \cdot K}$$

B) Cumulative Trotter Blocks Across All Steps ($N_{\text{Trotter, tot}}$):
Since step k contains k Trotter blocks ($\sum_{k=0}^{K-1} k = \frac{K(K-1)}{2}$):

$$\mathbf{N_{\text{Trotter, tot}} = 2 \cdot M \cdot \frac{K(K-1)}{2} = M \cdot K(K-1)}$$

C) Total Hardware Measurement Shots ($N_{\text{shots, tot}}$):

$$\mathbf{N_{\text{shots, tot}} = S \cdot N_{\text{circuits}} = 2 \cdot S \cdot M \cdot K}$$

2. Resource Scaling: $K = 50, S = 1\,000$ vs. Baseline $K = 300, S = 1\,024$

Metric	NISQ Regime ($K = 50$ Steps, $S = 1\,000$ Shots)			Baseline Regime ($K = 300$ Steps, $S = 1\,024$ Shots)		
	Mol. C ($N = 2$)	**Mol. A or B** ($N = 3$, each)	**All 3 Combined** (C + A + B)	**Mol. C** ($N = 2$)	**Mol. A or B** ($N = 3$, each)	**All 3 Combined** (C + A + B)
Symmetry States M_{sym} ($m_x > 0$)	**1** (++)	**4** (+ + +, . . .)	**9** states tot.	1	4 (each)	9 states tot.
Full Basis States M_{full} (2^N)	4	8	20 states tot.	4	8 (each)	20 states tot.
Time Steps K × Bases (X, Y)	50×2	50×2	50×2	300×2	300×2	300×2
Circuits N_{circuits} (Symmetry)	**100**	**400**	**900**	**600**	**2 400**	**5 400**
Circuits N_{circuits} (Full 2^N)	400	800	2 000	2 400	4 800	12 000
Trotter Blocks $N_{\text{Trotter, tot}}$ (Sym.)	**2 450**	**9 800**	**22 050**	**89 700**	**358 800**	**807 300**
Max 2Q Depth at Step $K-1$	**294** CZ	**882** CZ	max. 882 CZ	1 794 CZ	5 382 CZ	max. 5 382 CZ
Total Shots $N_{\text{shots, tot}}$ (Sym.)	**100 000**	**400 000**	**900 000**	**614 400**	**2.46** M	**5.53** M
Total Shots $N_{\text{shots, tot}}$ (Full 2^N)	400 000	800 000	2 000 000	2.46 M	4.92 M	12.29 M
QPU Time on `ibm_marrakesh` (Sym.)	**≈ 26** s	**≈ 112** s (**1.9** m)	**≈ 4.2** min	≈ 3.2 min	≈ 17.6 min	≈ 38.4 min
QPU Time on `ibm_marrakesh` (Full 2^N)	≈ 104 s	≈ 224 s (3.7 m)	≈ 9.2 min	≈ 12.8 min	≈ 35.2 min	≈ 83.2 min

3. Hardware Execution Time Breakdown on `ibm_marrakesh` ($K = 50, S = 1\,000$)

On the 156-qubit Heron r2 processor `ibm_marrakesh`, each shot comprises unitary gate execution ($t_{CZ} \approx 68$ ns per 2-qubit `CZ` gate; 6 CZ/step for $N = 2$, 18 CZ/step for $N = 3$) plus readout and active reset delay ($t_{\text{rep}} \approx 250\,\mu$s/shot):

Step-by-Step Calculation for $K = 50$ Steps ($\bar{k} = 24.5$ avg. steps) and $S = 1\,000$ Shots (Symmetry-Reduced $m_x > 0$):

- **Molecule C ($N = 2$ Spins, $M = 1$):** $N_{\text{circuits}} = 2 \times 1 \times 50 = \mathbf{100}$ circuits $\rightarrow$ $N_{\text{shots, tot}} = \mathbf{100\,000}$ shots.
  Avg. circuit time: $24.5 \times 6 \times 68$ ns $\approx 10\,\mu$s (gates) $+250\,\mu$s (readout/reset) $= 260\,\mu$s/shot $\implies$ $100\,000 \times 260\,\mu\text{s} \approx \mathbf{26}$ seconds.
- **Molecule A ($N = 3$ Spins, $M = 4$):** $N_{\text{circuits}} = 2 \times 4 \times 50 = \mathbf{400}$ circuits $\rightarrow$ $N_{\text{shots, tot}} = \mathbf{400\,000}$ shots.
  Avg. circuit time: $24.5 \times 18 \times 68$ ns $\approx 30\,\mu$s (gates) $+250\,\mu$s (readout/reset) $= 280\,\mu$s/shot $\implies$ $400\,000 \times 280\,\mu\text{s} \approx \mathbf{112}$ seconds (**1.9** min).
- **Molecule B ($N = 3$ Spins, $M = 4$):** Identical scaling to Molecule A: **400** circuits, **400 000** shots $\implies$ **112** seconds (**1.9** min).
- **All 3 Molecules Combined (A + B + C):** **900** circuits, **22 050** Trotter blocks, **900 000** shots $\implies$ **250** seconds ($\approx$ **4.2** minutes total QPU time).

Noise Physics & Signal Processing in Quantum NMR Simulations

Qiskit Fall Fest 2026 – HQS Challenge | Classical Apodization vs. Hardware Gate Decoherence vs. Statistical Shot Noise

Characteristic	1. Classical Apodization (`noise_strength` Γ) (Post-Processing Exponential Window)	2. Hardware Gate Noise (Γ_{dep}) (Depolarizing Error & Qubit Decoherence T_1, T_2)	3. Statistical Shot Noise ($S = N_{\text{shots}}$) (Finite-Sample Projection Noise at Readout)
Physical / Algorithmic Origin	Deterministic post-processing filter applied **after** quantum simulation prior to FFT to suppress truncation ringing and model natural T_2^* relaxation.	Physical decoherence occurring **during** circuit execution as 1-qubit and 2-qubit gates accumulate depolarizing errors across k Trotter steps.	Statistical estimation error arising **at terminal readout** when averaging S binary projective measurements ($\{0,1\}^N$) per time step.
Time-Domain Effect on FID Signal $C(t)$	Multiplies the ideal correlation function by a smooth exponential decay envelope: $$C_{\text{window}}(t) = C_{\text{ideal}}(t) \cdot e^{-\Gamma t}$$	Attenuates transverse magnetization with circuit depth k, acting as an exponential damping factor: $$C_{\text{dep}}(t) \approx C_{\text{ideal}}(t) \cdot e^{-4\Gamma_{\text{dep}} t}$$	Superimposes uncorrelated zero-mean random fluctuations $\epsilon(t_k)$ of standard deviation $\sigma \propto 1/\sqrt{S}$: $$C_{\text{shot}}(t_k) = C_{\text{ideal}}(t_k) + \epsilon(t_k), \quad \sigma \approx \frac{\sqrt{N}}{2\sqrt{S}}$$
Frequency-Domain Effect on Spectrum $S(\omega)$	**Homogeneous Line Broadening:** Convolves sharp delta lines into smooth Lorentzian peaks of Full Width at Half Maximum: $$\Delta f_{\text{FWHM}} = \frac{\Gamma}{\pi} \text{ Hz}$$	**Intrinsic Spectral Broadening:** Broadens Lorentzian resonance peaks without shifting their center frequencies: $$\Delta f_{\text{FWHM, QPU}} = \frac{4\Gamma_{\text{dep}}}{\pi} \text{ Hz}$$	**Spectral Baseline Noise Floor:** Does **not** broaden peaks; introduces stochastic white-noise fluctuations across the entire baseline (SNR $\propto \sqrt{S}$).
Resolution Limit & Failure Mode	Multiplet splittings merge into a single broad peak when $\Delta f_{\text{FWHM}} \gtrsim \lvert J_{k\ell} \rvert$, i.e., above $\Gamma_{\max} \approx \pi \cdot \lvert J_{k\ell} \rvert$.	Signal decays to zero after $K_{\max} \approx \frac{1.8}{n_{2Q} p_{2Q}}$ steps, erasing fine J-coupling splittings.	Low-intensity peaks are submerged beneath the baseline noise floor if S is too small ($S < 256$).
Qiskit Simulation Method	`window = np.exp(-noise_strength * times)` `c_xx = c_xx * window`	`AerSimulator(noise_model=...)` with `depolarizing_error(p, 2)` on 2-qubit gates.	`Estimator(default_precision=1/np.sqrt(S))` or `Sampler(shots=1024)`.

Intrinsic Noise Tolerance of Quantum NMR Spectroscopy:
Depolarizing gate noise on NISQ processors ($e^{-4\Gamma_{\text{dep}} t}$) and natural transverse spin-spin relaxation in liquid-state NMR (e^{-t/T_2^*}) exhibit the **exact same mathematical form** as the exponential apodization filter $e^{-\Gamma t}$. Consequently, moderate quantum hardware decoherence does not corrupt peak positions; it naturally manifests as Lorentzian line broadening, preserving full analytical utility as long as $4\Gamma_{\text{dep}} < \pi \cdot \lvert J_{\min} \rvert$.

QPU Benchmark: `ibm_kingston` vs. `ibm_marrakesh` (156-Qubit Heron r2)

Qiskit Fall Fest 2026 – HQS Challenge | Qubit Calibration Metrics, Gate Error Scaling $\mathcal{F}(K)$, Step Selection & Experimental Spectra

1. Hardware Analysis & Optimization Workflow

- **Symmetry ($m_x > 0$):** Cuts circuits by $4\times$ ($N = 2 : M = 1$) and $2\times$ ($N = 3 : M = 4$).
- **Calibration Graph Search (`backend.target`):** Picks linear chain Q minimizing:

$$\text{Cost}(Q) = N_{\text{CZ}} \sum_{(u,v)\in Q} \epsilon_{\text{CZ}}(u,v) + \sum_{q\in Q} \epsilon_{\text{RO}}(q) + \frac{w_{T_2}}{\bar{T}_2(Q)}$$

- **Level-3 + KAK Synthesis:** 0 SWAPs; $N = 2 \to$ **3** CZ ($\forall k$); $N = 3 \to$ **13.2** CZ/step.
- **Runtime:** `XpXm` Dynamical Decoupling + `resilience_level=0` + parallel batching.

2. Live Qubit Error Comparison (`kingston` vs. `marrakesh`)

Hardware Metric / Layout	`ibm_kingston` (Winner)		`ibm_marrakesh`	
	2Q: $\mathbf{Q}[1,2]$	**3Q:** $\mathbf{Q}[1,2,3]$	**2Q:** $\mathbf{Q}[147,148]$	**3Q:** $\mathbf{Q}[5,6,7]$
Chip Median ϵ_{CZ} / ϵ_{RO}	0.181% / 1.07%		0.237% / 1.66%	
Chip Median T_1 / T_2	$254.3\,\mu$s / $142.5\,\mu$s		$163.7\,\mu$s / $92.9\,\mu$s	
Pair $\epsilon_{\text{CZ}}(q_0,q_1)$ / (q_1,q_2)	$\mathbf{0.079\%}$ / —	$\mathbf{0.079\%}$ / $\mathbf{0.116\%}$	0.116% / —	0.129% / 0.156%
Mean 2Q Error $\bar{\epsilon}_{\text{CZ}}$	$\mathbf{0.079\%}$	$\mathbf{0.097\%}$ (-32%)	0.116%	0.142%
Readout $\epsilon_{\text{RO}}(q_0,q_1,q_2)$	$0.39, 1.32\%$	$0.39, 1.32, 0.98\%$	$0.63, 0.49\%$	$0.24, 1.22, 2.39\%$
Mean Readout $\bar{\epsilon}_{\text{RO}}$	0.85%	$\mathbf{0.90\%}$ (-30%)	$\mathbf{0.56\%}$	1.28%
Chain Mean T_2	$312.4\,\mu$s	$\mathbf{338.1}\,\mu$s $(+36\%)$	$122.7\,\mu$s	$248.7\,\mu$s

3. Error Calculation, Step Impact & Selection of K ($\Delta t = 2.5$ ms)

Gate Survival Fidelity at Step k ($N_{\text{CZ}}(k) \approx 13.2 \cdot k$ for $N = 3$, 3 for $N = 2$):

$$\mathcal{F}_{\text{gate}}(k) = (1-\bar{\epsilon}_{\text{CZ}})^{N_{\text{CZ}}(k)} \approx e^{-\bar{\epsilon}_{\text{CZ}} N_{\text{CZ}}(k)}, \qquad \Delta f_{\text{FID}} = \frac{1}{K\cdot\Delta t} \text{ vs. } \Delta f_{\text{QPU}} = \frac{4\Gamma_{\text{dep}}}{\pi}$$

- **Mol. C ($N = 2 \to K = 80$ / 60):** KAK synthesis fixes $N_{\text{CZ}}(k) = \mathbf{3}$ CZ $\forall k \geq 1$:

  $\mathcal{F}^{\text{King}}_{\text{gate}} = (1-0.00079)^3 = \mathbf{99.76\%}$, $\mathcal{F}^{\text{Marr}}_{\text{gate}} = (1-0.00116)^3 = \mathbf{99.65\%}$

  $\Longrightarrow$ Depth-independent! High $K = 60$–80 yields sharp doublets ($\Delta f_{\text{FID}} = 5.0$–$6.7$ Hz) in 34–46 s.
- **Mol. A & B on `marrakesh` ($N = 3 \to K = 45$, 594 CZ):**

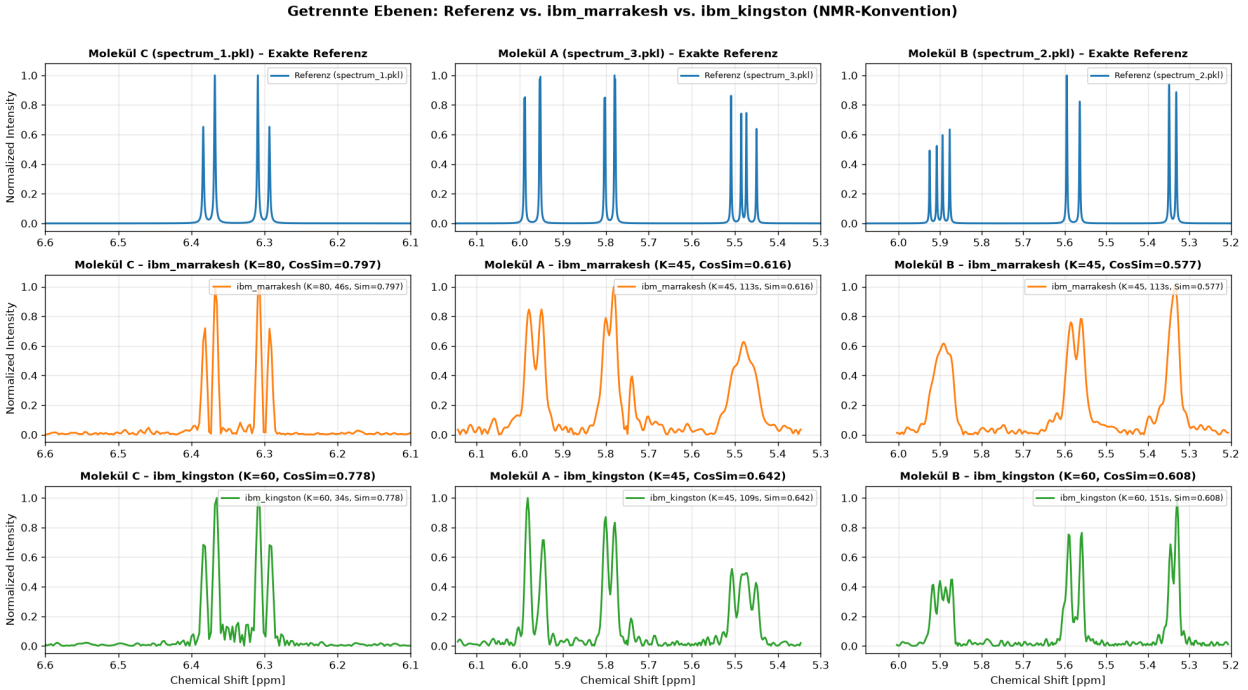
  $\mathcal{F}^{\text{Marr}}_{\text{gate}}(45) = (1-0.00142)^{594} = \mathbf{43.0\%}$ $\left(\text{at } K = 60, 795 \text{ CZ} : (1-0.00142)^{795} = \mathbf{32.3\%} \to \text{noise}\right)$

  $\Longrightarrow$ $K = 45$ ($\Delta f_{\text{FID}} = 8.9$ Hz, 113 s) avoids tail decoherence; resolves doublets, merges B's quadruplet.
- **Mol. B on `kingston` ($N = 3 \to K = 60$, 795 CZ) & Mol. A ($K = 45$, 594 CZ):**

  $\mathcal{F}^{\text{King}}_{\text{gate}}(60) = (1-0.00097)^{795} = \mathbf{46.2\%} > \mathcal{F}^{\text{Marr}}_{\text{gate}}(45)$, $\mathcal{F}^{\text{King}}_{\text{gate}}(45) = (1-0.00097)^{594} = \mathbf{56.2\%}$

  $\Longrightarrow$ 32% lower $\bar{\epsilon}_{\text{CZ}}$ allows $K = 60$ ($\Delta f_{\text{FID}} = \mathbf{6.67}$ Hz $< J_{\text{AB}}$), resolving **all 4 quadruplet peaks**!

4. Experimental QPU Spectra (`Molecule_Kingston_Marrakesh`)



5. Plot & Run Parameters ($S = 1\,000$ Shots/Circuit, $\Gamma_{\text{apod}} = 3.5$ s^{-1})

Molecule	Backend	Qubits	Steps K	Circuits	Max CZ	$\mathcal{F}_{\text{gate}}(K)$	Time / CosSim
Mol. C ($N = 2$)	`marrakesh`	$Q[147,148]$	$K = 80$	160	3 CZ	99.65%	46 s / **0.797**
`spectrum_1`	`kingston`	$Q[1,2]$	$K = 60$	120	3 CZ	$\mathbf{99.76\%}$	**34** s / 0.778
Mol. A ($N = 3$)	`marrakesh`	$Q[5,6,7]$	$K = 45$	360	594 CZ	43.0%	113 s / 0.616
`spectrum_3`	`kingston`	$Q[1,2,3]$	$K = 45$	360	594 CZ	$\mathbf{56.2\%}$	**109** s / **0.642**
Mol. B ($N = 3$)	`marrakesh`	$Q[5,6,7]$	$K = 45$	360	594 CZ	43.0%	113 s / 0.577
`spectrum_2`	`kingston`	$Q[1,2,3]$	$\mathbf{K = 60}$	480	795 CZ	$\mathbf{46.2\%}$	151 s / **0.608**